

● HARD

● EXPRESSION OF IDEAS

Expression of Ideas

10 questions · ~15 min

08

Q1

EXPRESSION OF IDEAS

Economist Elinor Ostrom's studies of communities around the world have empirically demonstrated that common pool resources, such as grazing lands, can be sustainably managed by the people who use them (rather than through private entities or centralized governments). _____blank Ostrom's work is a repudiation of the "tragedy of the commons," the view that individuals will inevitably overexploit a finite shared resource if given unfettered access to it.

Which choice completes the text with the most logical transition?

- A. By contrast,
- B. For example,
- C. That said,
- D. As such,

While researching a topic, a student has taken the following notes:

- The background colors of US and UK road signs are used to denote each sign's purpose.
- Directional signs are a type of sign containing information such as route names, distance to a destination, etc.
- Highways, major roadways, and minor roadways in the US generally use green for directional signs.
- Highways in the UK generally use blue for directional signs.
- Major roadways in the UK generally use green for directional signs.

The student wants to contrast how green backgrounds are used in US and UK road signs. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Blue backgrounds are generally used on directional highway signs in the UK, in contrast to the green highway signs used in the US.
- B. In the US, green signs indicate directional information on both highways and major roadways, but in the UK, directional signs of this color generally appear only on major roadways.
- C. Green road signs are used in both the US and UK to denote directional information on major roadways, like distance to a destination or route names.
- D. Both the UK and the US use directional signs, which include information on route names and distance to a destination.

Seismologists Kaiqing Yuan and Barbara Romanowicz have proposed that the magma fueling Iceland's more than 30 active volcano systems emerges from deep within Earth. The great depths involved—nearly 3,000 km—mark Iceland's volcanoes as extreme outliers; _____ blank many of Earth's volcanoes are fed by shallow pockets of magma found less than 15 km below the surface.

Which choice completes the text with the most logical transition?

- A. indeed,
- B. nevertheless,
- C. in addition,
- D. consequently,

While researching a topic, a student has taken the following notes:

- In 1971, experimental musician Pauline Oliveros created *Sonic Meditations*.
- *Sonic Meditations* is not music but rather a series of sound-based exercises called meditations.
- Each meditation consists of instructions for participants to make, imagine, listen to, or remember sounds.
- The instructions for Meditation V state, “walk so silently that the bottoms of your feet become ears.”
- Those for Meditation XVIII state, “listen to a sound until you no longer recognize it.”

The student wants to provide an explanation and an example of Oliveros’s *Sonic Meditations*. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. *Sonic Meditations* is not music but rather a series of sound-based meditations that consist of instructions; Meditation XVIII, for instance, instructs participants to “listen to a sound until you no longer recognize it.”
- B. In 1971, Oliveros created *Sonic Meditations*, a series of meditations that consist of instructions for participants to make, imagine, listen to, or remember sounds.
- C. “Walk so silently that the bottoms of your feet become ears” is one example of the instructions found in Oliveros’s *Sonic Meditations*.
- D. While both meditations consist of instructions, Meditation XVIII instructs participants to “listen,” whereas Meditation V instructs participants to “walk.”

If the formation of Earth’s mantle had been purely a product of core differentiation—whereby heavier elements sink toward the core and lighter elements rise—the upper mantle would be depleted of heavy siderophile elements. Siderophiles are much more abundant in the mantle than predicted in that model, however.

_____blank extraterrestrial material containing siderophiles, likely from asteroid or comet impacts, almost certainly accreted to Earth following core differentiation.

Which choice completes the text with the most logical transition?

- A. That said,
- B. Hence,
- C. For example,
- D. Likewise,

While researching a topic, a student has taken the following notes:

- The ancient Arab dhow was a sailing vessel distinguishable by its triangular sails and stitched hull construction.
- Dhows were used primarily for trade along the coasts of Arab, South Asian, and East African countries.
- Contemporary shipbuilders in Oman use a mix of modern and traditional materials to build replicas of ancient dhows.
- Most of the materials used are traditional.
- Replica hulls are stitched together using the same traditional coconut palm fiber rope used on the hulls of ancient dhows.

The student wants to make a generalization about the materials used in dhow replicas. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. A traditional material that was used to stitch together the hulls of ancient dhows, coconut palm fiber rope is still used by shipbuilders.
- B. The ancient Arab dhow was a sailing vessel used primarily for trade and distinguishable by its triangular sails.
- C. Although most materials used in dhow replicas are traditional, some modern materials are used.
- D. Contemporary shipbuilders in Oman build replicas of the dhow, which was an ancient sailing vessel with a stitched hull construction.

Magnetic levitation (maglev) trains are suspended above a track by powerful electromagnets, reducing friction and thus allowing for much faster speeds. Though maglev advocates in the US have long imagined these trains crisscrossing the country, their dream remains unrealized. _____ blank of the handful of maglev trains currently in operation, all are in Asia.

Which choice completes the text with the most logical transition?

- A. In fact,
- B. To that end,
- C. Nevertheless,
- D. That said,

While researching a topic, a student has taken the following notes:

- The popular wood-wide web theory posits that trees can communicate and exchange resources with one another via common mycorrhizal networks (CMNs) of fungi.
- Ecologist Dr. Suzanne Simard first suggested this theory in 1997.
- She described trees as “super-cooperators.”
- In the 2022 study “The Decay of the Wood-Wide Web?,” mycologist Dr. Justine Karst and colleagues evaluated dozens of CMN studies.
- They write that CMNs “have captured the interest of broad audiences. We are concerned, however, that recent claims about CMNs in forests are disconnected from evidence.”

The student wants to use a quotation to emphasize a potential problem with the wood-wide web theory. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Describing trees as “super-cooperators,” Simard first suggested that trees can exchange resources with one another in 1997.
- B. In “The Decay of the Wood-Wide Web?,” Karst and colleagues note that common mycorrhizal networks “have captured the interest of broad audiences.”
- C. After evaluating dozens of CMN studies, Karst and colleagues expressed concern that recent claims about common mycorrhizal networks are “disconnected from evidence.”
- D. Despite the concerns expressed in the 2022 study “The Decay of the Wood-Wide Web?,” the wood-wide web theory remains popular.

While researching a topic, a student has taken the following notes:

- Two opposing theories of vision divided scholars for many centuries.
- The ancient Greek mathematician Euclid (circa 300 BCE) supported the extramission theory.
- This theory held that the eyes emit a form of radiation that illuminates objects in its range.
- The ancient Greek philosopher Aristotle (384–322 BCE) supported the intromission theory.
- This theory held that objects emit a form of radiation that reaches the eyes.
- In the eleventh century, Arab mathematician Ibn al-Haytham (965–1040 CE) largely settled the debate with the first conclusive experiments supporting intromission.

The student wants to provide a historical overview of the two theories. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Scholars were divided between the extramission and intromission theories of vision until Ibn al-Haytham's eleventh-century experiments largely settled the debate in support of intromission.
- B. Through two opposing theories of vision—extramission and intromission—Euclid, Aristotle, and Ibn al-Haytham held that a form of radiation is emitted either from objects or from the eyes.
- C. While Ibn al-Haytham largely settled the debate in the eleventh century, Aristotle supported the theory of intromission centuries before.
- D. Before the eleventh century, the ancient Greek philosopher Aristotle supported the intromission theory, which held that objects emit a form of radiation that reaches the eyes.

While researching a topic, a student has taken the following notes:

- Ulaanbaatar is the capital of Mongolia.
- The city's population is 907,802.
- Ulaanbaatar contains 31.98 percent of Mongolia's population.
- Hanoi is the capital of Vietnam.
- The city's population is 7,781,631.
- Hanoi contains 8.14 percent of Vietnam's population.

The student wants to emphasize the relative sizes of the two capitals' populations. Which choice most effectively uses information from the given sentences to emphasize the relative sizes of the two capitals' populations?

- A. Mongolia's capital is Ulaanbaatar, which has 907,802 people, and Vietnam's capital is Hanoi, which has 7,781,631 people.
- B. Comparing Vietnam and Mongolia, 7,781,631 is 8.14 percent of Vietnam's population, and 907,802 is 31.98 percent of Mongolia's.
- C. Even though Hanoi (population 7,781,631) is larger than Ulaanbaatar (population 907,802), Ulaanbaatar accounts for more of its country's population.
- D. The populations of the capitals of Mongolia and Vietnam are 907,802 (Ulaanbaatar) and 7,781,631 (Hanoi), respectively.